

Quinten Bubberman

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Third-year game programming student at Breda University of Applied Sciences specializing in graphics and engine systems programming. Experienced with Vulkan rendering pipelines, ECS architecture, and performance optimization. Focused on low-level systems, GPU-driven techniques, and building robust engine tools.

Skills

Programming languages: C++, Rust, Python, GLSL

APIs: Vulkan, EnTT, SDL3, GLFW

Engines: Unreal Engine, Godot Engine

Tools: Git, Perforce, Rider, Visual Studio, NSight Graphics, Helix, Linux

Natural Languages: Dutch (native), English (fluent)

Projects

Vulkan GPU-Driven Renderer

Nov 2025 - Present

C++ Graphics Programmer - Solo

Implemented Multi-Draw Indirect with compute shader generating draw commands from storage buffers, rendering 636K objects at 60 FPS in single draw call. Built CPU frustum culling with plane extraction from projection matrix, achieving 5x performance improvement.

Beetle: Custom Engine

Feb 2025 – Apr 2025

C++ Engine & Tools Programmer - Team

Developed Blender pipeline with custom GLTF property exporting and Python tooling. Built ECS-based UI system with live editor, cubic Bezier spline system for AI pathfinding, spring arm camera, 3D audio with FMOD, and input mapping. Shipped on PC and Steam Deck.

Vulkan Renderer Replacement

Nov 2024 – Jan 2025

C++ Graphics Programmer - Solo

Migrated existing engine rendering backend from OpenGL to Vulkan, implementing PBR with IBL and instanced rendering. Built shadow mapping for directional lights, MSAA, and post-processing pipeline. Profiled with NSight Graphics and optimized vertex shader I/O and buffer transfers, reducing frame time by 33%.

Education

Breda University of Applied Sciences

Bachelor's, Game Technologies

Nehalennia

HAVO (Higher General Secondary Education)

Sept 2023 – Jul 2027

Breda, Netherlands

Sept 2018 – Jul 2023

Middelburg, Netherlands